Model Agnostic monocular depth estimation with lightweight lidar array

16-663 F1Tenth Autonomous Racing | Spring 2026

Depth from single images are up to scale

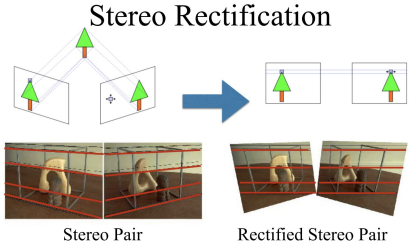
- **Projection matrix is non-invertible.**
  - Ambiguity for scale

- **We can recover from**
  - **A precise lidar**
  - Stereo + Features - degrades when the scene is feature-less
  - SfM - baseline is the movement, too small = error is large.
    - F1 cars move too quick for features to stay in scene

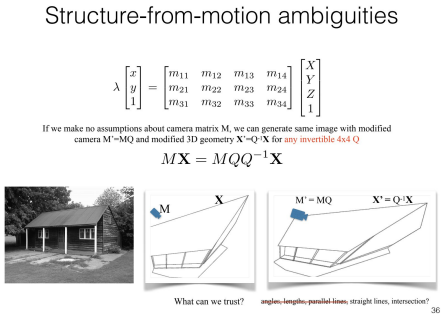
Our approach: use 1D LiDAR array for precise, low-cost metric scale recovery

Related Works

Solution 1: Stereo

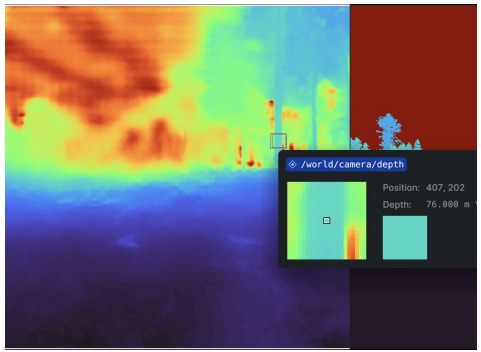


Solution 2: SfM + IMU



Solution 3: Learning Based





Source: 16-820 Advanced Computer Vision

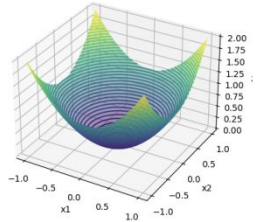
Recall that PSD matrices can be visualized as quadratic forms

$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} B \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where B is

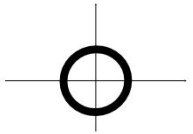
$$B = A^T A = V \Sigma^2 V^T = \begin{vmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{vmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} -\mathbf{v}_1^T - \\ -\mathbf{v}_2^T - \end{bmatrix}$$

$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + x_2^2$$

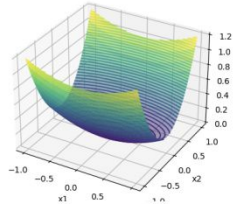


2D visual:

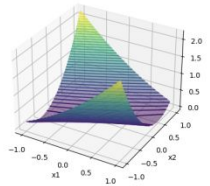
$f(x_1, x_2) = 1$

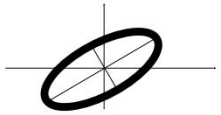


$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda_1 x_1^2 + \lambda_2 x_2^2$$



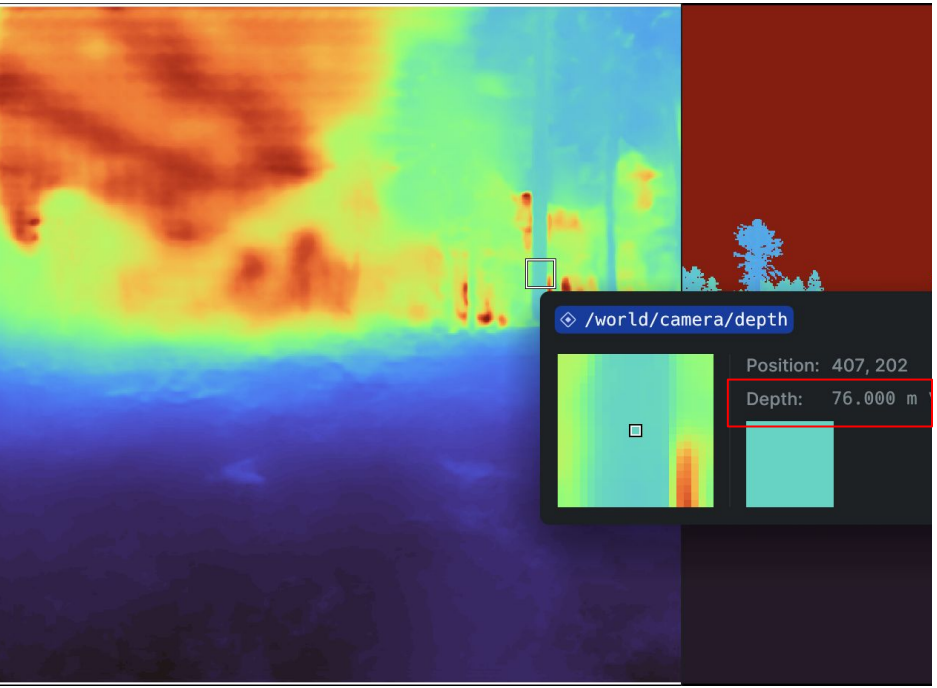
$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{vmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{vmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} -\mathbf{v}_1^T - \\ -\mathbf{v}_2^T - \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

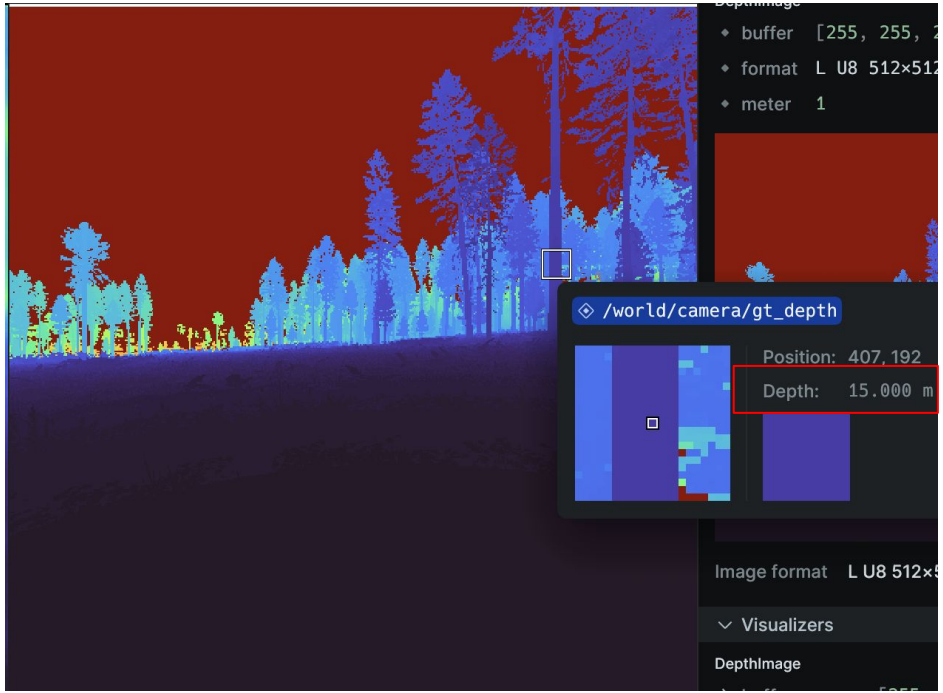




$$= \lambda_1 \hat{x}_1^2 + \lambda_2 \hat{x}_2^2, \quad \text{where} \quad \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} -\mathbf{v}_1^T - \\ -\mathbf{v}_2^T - \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution 3-3: (Absolute) Depth Models - Domain Gap





Proposed Approach

Our Approach

1. Sparse Lidar Scan to provide absolute + Dense Model prediction to provide relative

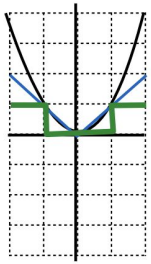


RANSAC

2. RANSAC for outlier removal

What's really going on: robust losses (robust statistics)

https://en.wikipedia.org/wiki/Robust_statistics



L2 error = $||x||^2$ penalizes outliers too much

L1 error = $|x|$ error: better

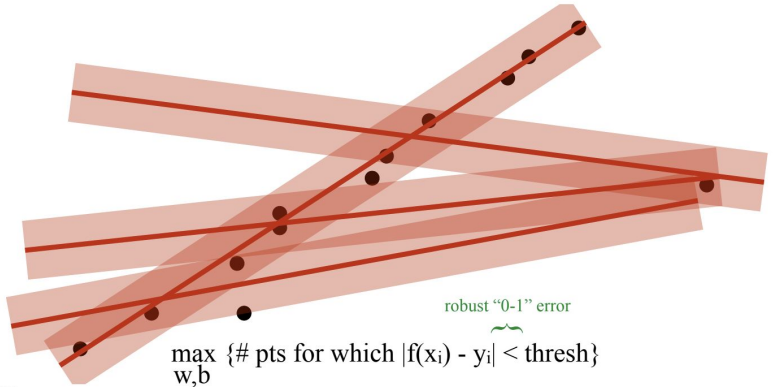
0-1 error = $\begin{cases} 0 & \text{if } |x| < \text{thresh} \\ 1 & \text{otherwise} \end{cases}$ "ideal" but difficult to optimize

One can view RANSAC as an algorithm for optimizing a *robust* error model

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RANSAC algorithm

https://en.wikipedia.org/wiki/Random_sample_consensus



robust "0-1" error

$$\max_{w,b} \{\#\text{ pts for which } |f(x_i) - y_i| < \text{thresh}\}$$

Key idea:

- (try to) Brute-force search for optimal solution with many "inliers"
- Exploit fact that inliers will all vote for same soln but outliers will vote in different ways (*stochastic* search rather than exhaustive search)

Source: 16-820 Advanced Computer Vision

How to measure Success

Method	Encoder	KITTI AbsRel	KITTI d1	NYUv2 AbsRel	NYUv2 d1	Sintel AbsRel	DDAD AbsRel	ETH3D AbsRel	DIODE AbsRel
MiDaS v3.1	ViT-L	0.127	0.850	0.048	0.980	0.587	0.251	0.139	0.075
Depth Any.	ViT-S	0.080	0.936	0.053	0.972	0.464	0.247	**0.127**	0.076
Depth Any.	ViT-B	0.080	0.939	0.046	0.979	0.432	0.232	**0.126**	0.069
Depth Any.	ViT-L	**0.076**	**0.947**	**0.043**	**0.981**	0.458	**0.230**	0.127	**0.066**

Model selection benchmarks (relative depth)

Our evaluation: metric depth accuracy

MAE

```
(1/n) * sum( |di - di*| )
```

Mean absolute error in meters.
Directly answers: how many meters off on average?

Target: < 0.15 m

RMSE

```
sqrt( (1/n) * sum( (di - di*)^2 ) )
```

Root mean squared error.
Penalizes large errors more — one bad pixel matters.

Target: < 0.25 m

AbsRel

```
(1/n) * sum( |di - di*| / di* )
```

Relative error, scale-aware.
Kept as secondary metric for comparison with benchmarks.

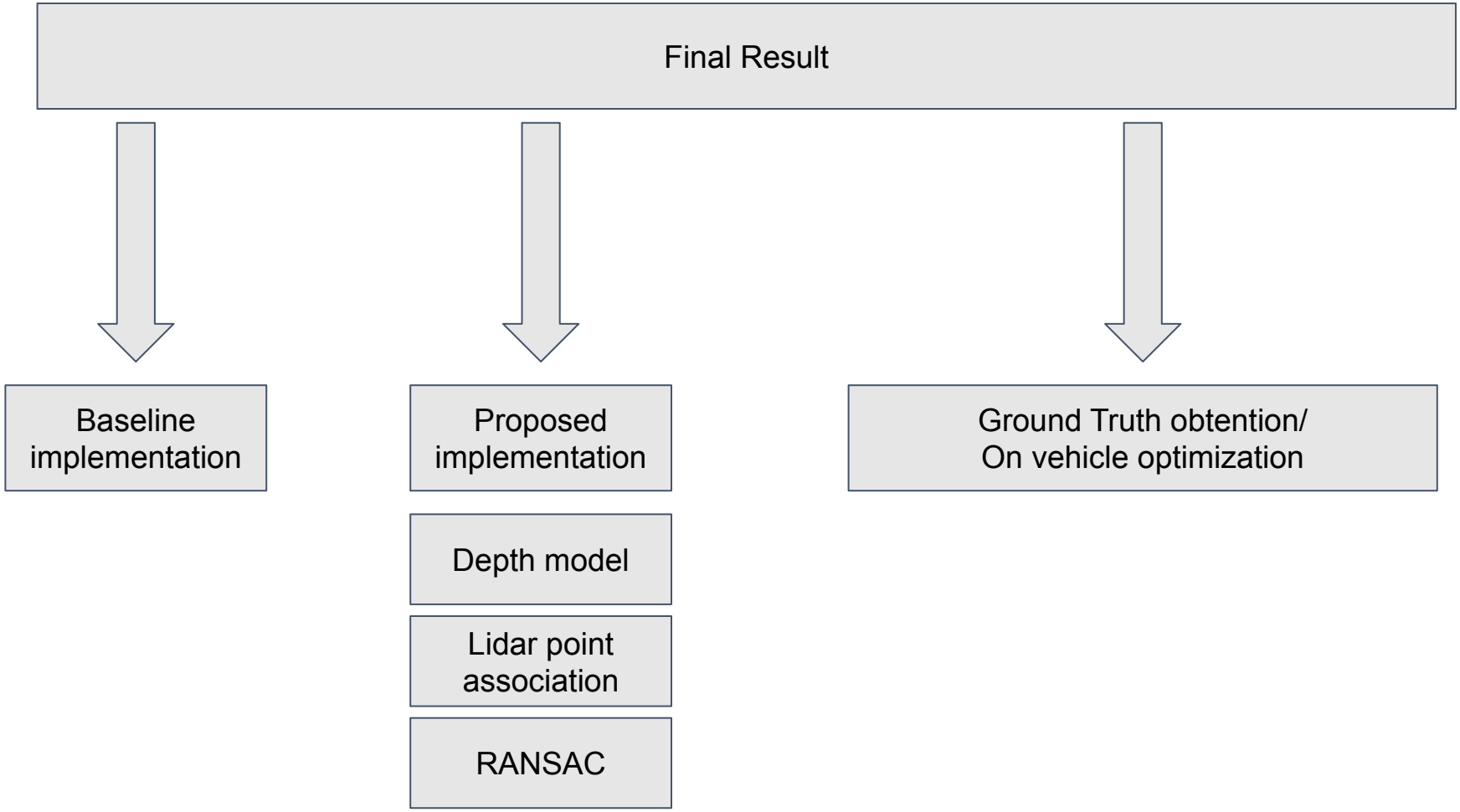
Target: < 15%

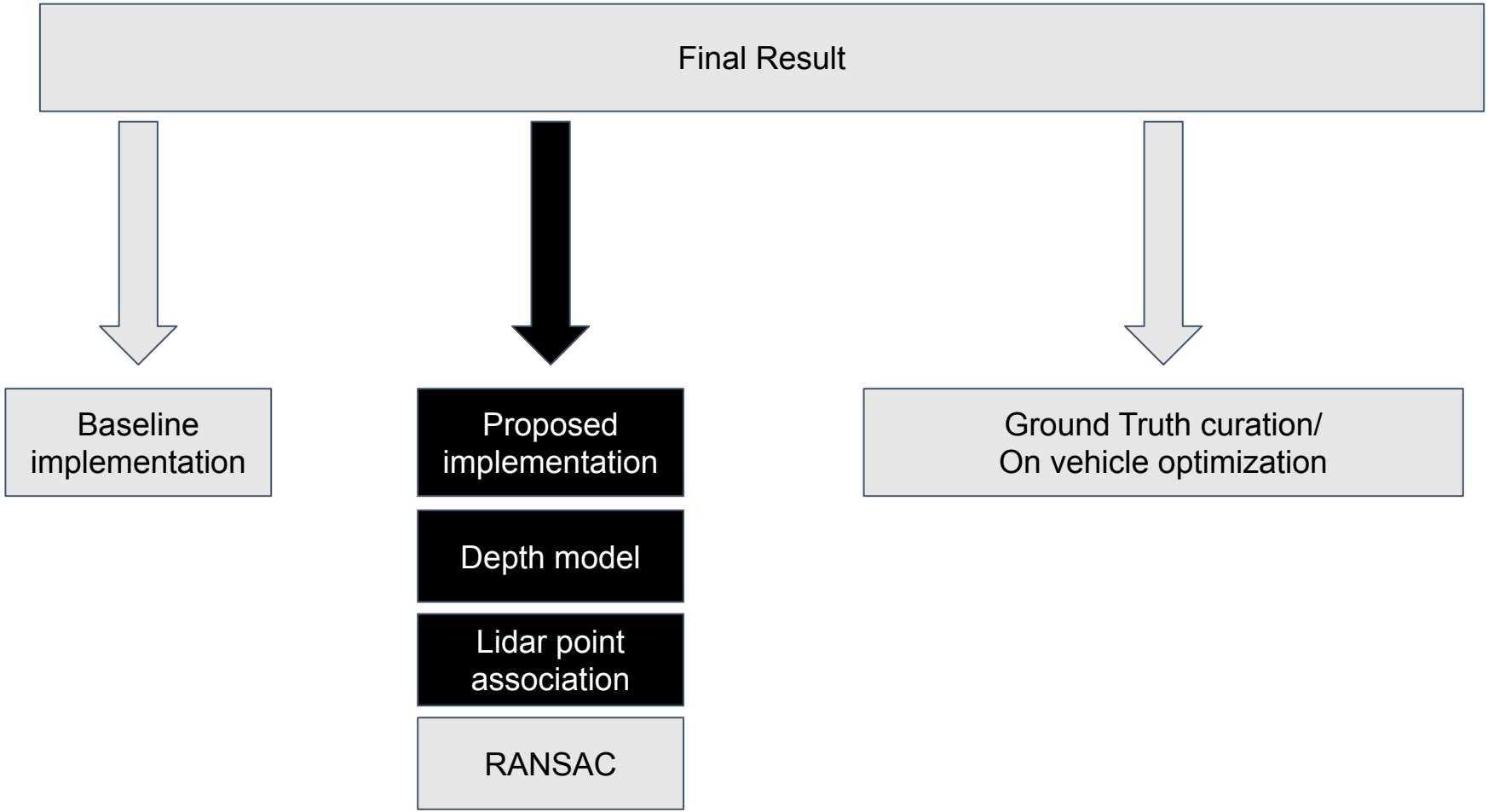
Ground truth: 2D LiDAR scans on the F1Tenth platform

Timeline

Simulation Pipeline Mar 24 – Apr 4	Benchmark Depth Anything V2 variants (Small/Large/Metric). Establish ground truth comparison framework.
1D LiDAR Fusion Apr 7 – 11	Implement recovery using sparse 1D LiDAR anchor points. Evaluate metric accuracy vs. ground truth 2D LiDAR scans.
Integration & Midstream Apr 14 – 18	Connect metric depth output to navigation stack (gap finder / pure pursuit). Present midstream results. Begin hardware deployment.
Hardware & Racing Apr 21 – 25	Deploy on physical F1Tenth car. Tune for real-time performance. Test in NSH Atrium. Race 3 preparation.
Demo & Report Apr 27 – May 1	Final demo with live car demonstration. Write and submit report.

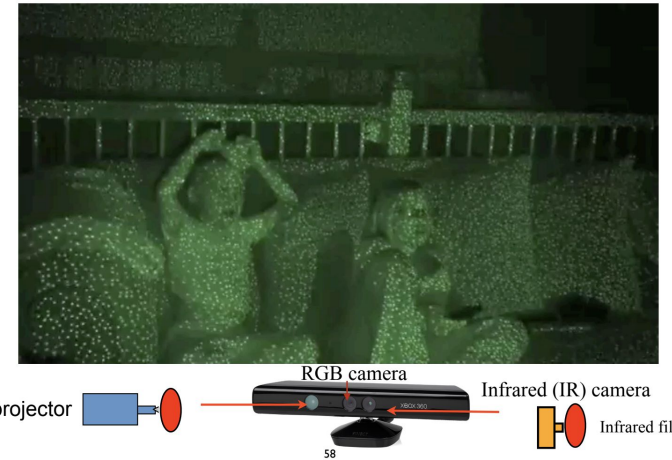
Current Progress


Final Result
Baseline implementation
Proposed implementation
Depth model
Lidar point association
RANSAC
Ground Truth obtention/ On vehicle optimization


Final Result
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Recap - Stereo & D435i



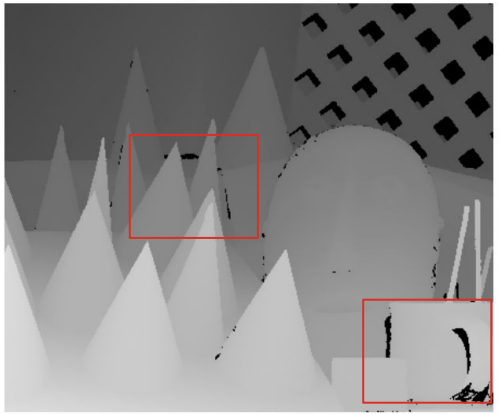


Source: 16-663 - L14 & 16820 L15

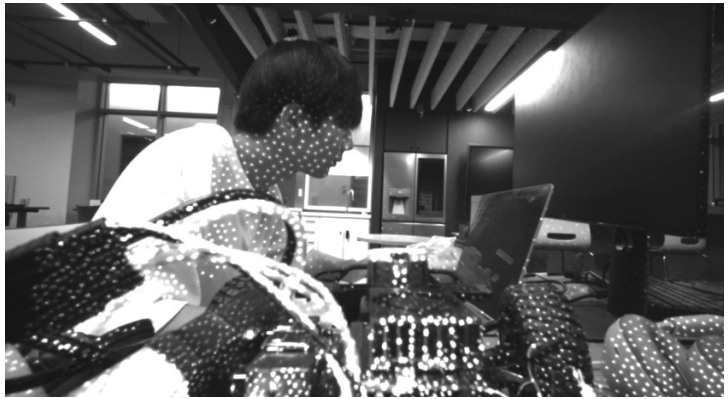




Disparity values (0-64)



Note how disparity is larger (brighter) for closer surfaces.

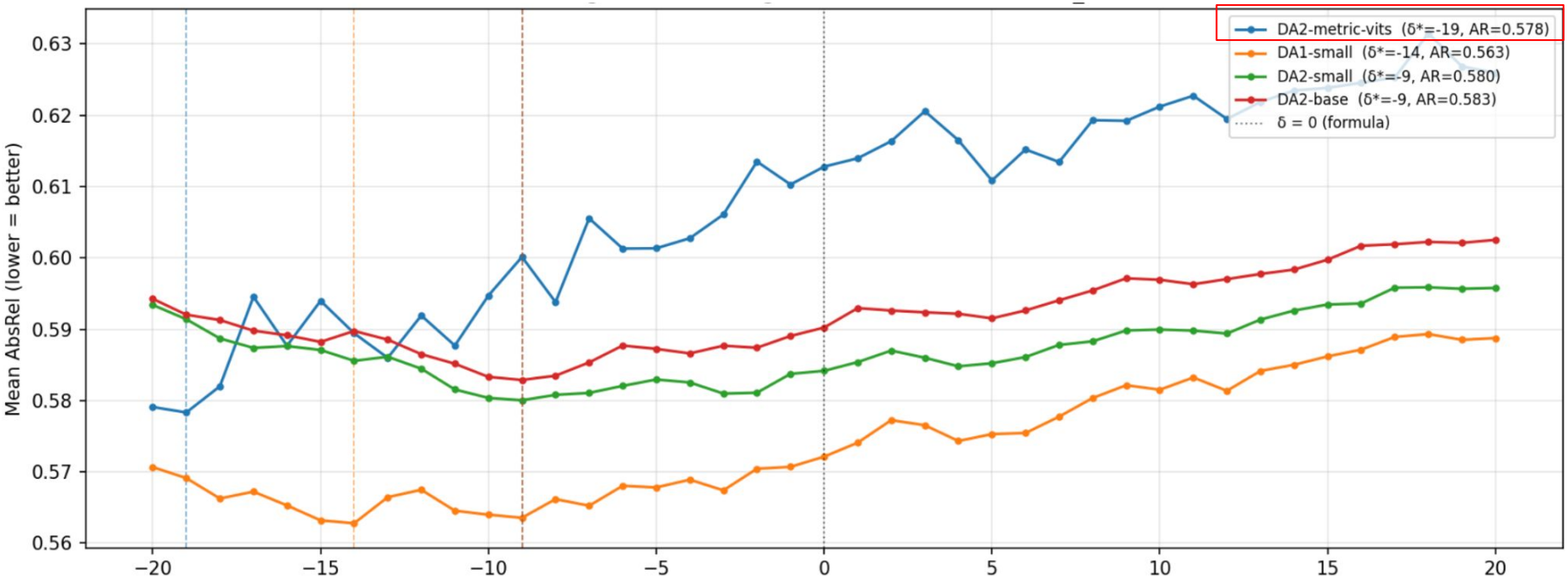


Depth model vs (pseudo) ground truth from D435i



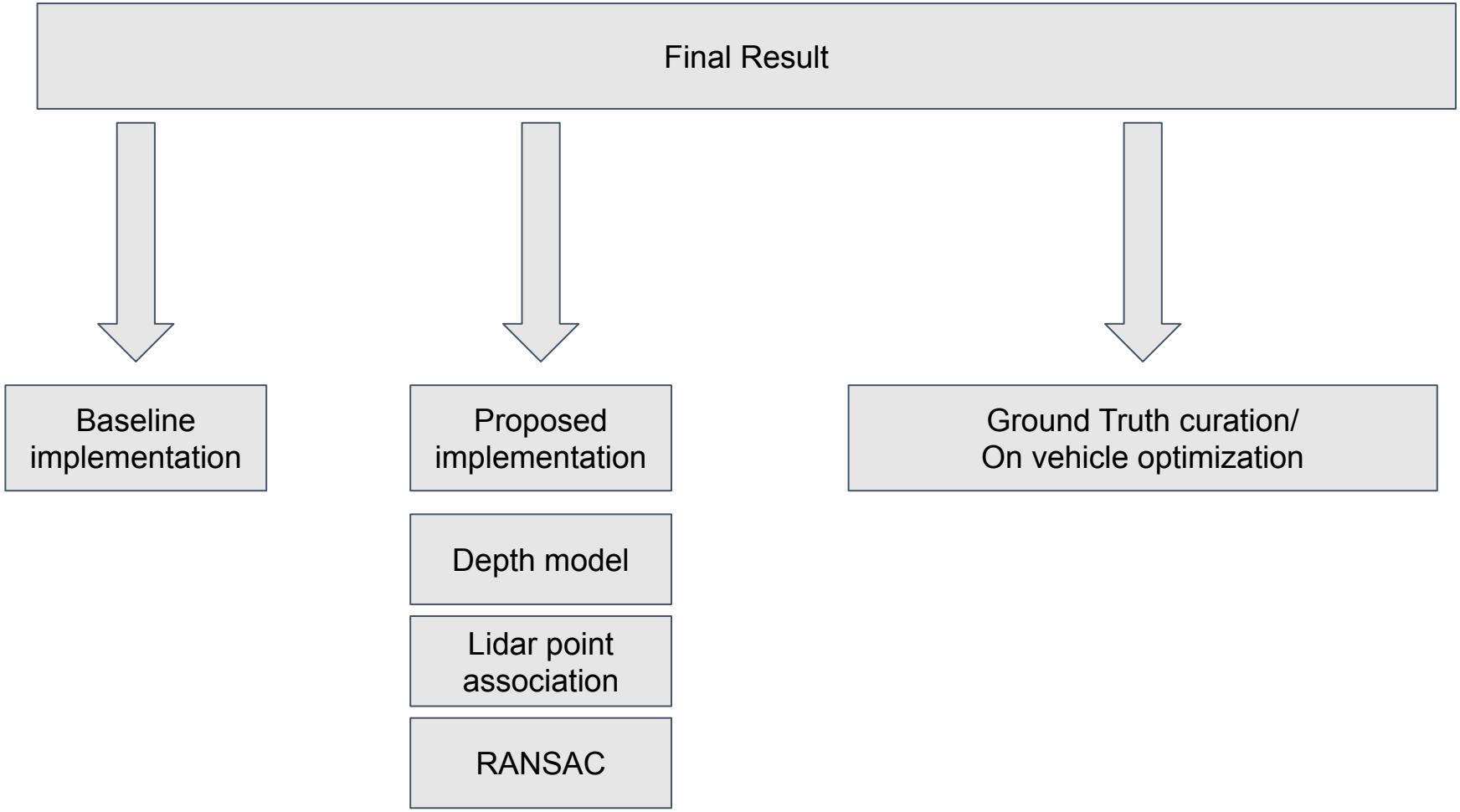


Lidar Points Association



Delta controls timestamp diff b/t lidar and rgb messages. Models should perform best when the lidar and rgb frames are perfectly aligned.

$$AbsRel = \frac{1}{n}\sum \frac{|d_i - d_i^*|}{d_i^*}$$


Final Result
Baseline implementation
Proposed implementation
Depth model
Lidar point association
RANSAC
Ground Truth curation/ On vehicle optimization